

ASK MY PDF BOT - README WITH OUTPUT

Project Overview

Ask My PDF Bot is an AI-powered PDF Question Answering system built using RAG (Retrieval-Augmented Generation), FAISS, Sentence Transformers, and Hugging Face Transformers. Users can upload a PDF file and ask one question related to the document.

Features

- Upload PDF files
- Extract text automatically
- Generate embeddings using Sentence Transformers
- Store vectors using FAISS
- Retrieve relevant context
- Generate AI-based answers using DistilGPT2
- Stops automatically after one answer

Technologies Used

Python, Google Colab, PyPDF, Sentence Transformers, FAISS, Transformers, DistilGPT2, Torch, NumPy

Installation Commands

```
!pip install -q transformers
!pip install -q sentence-transformers
!pip install -q faiss-cpu
!pip install -q pypdf
!pip install -q torch
```

Workflow

1. Upload PDF
2. Extract Text
3. Split into Chunks
4. Generate Embeddings
5. Create FAISS Index
6. Ask Question
7. Retrieve Similar Chunks
8. Generate AI Answer

Sample Output

PDF Uploaded: Moscow-Creative-Resume-Template (2).pdf

PDF Loaded Successfully

Total Chunks: 7

Embedding Model Loaded
Embeddings Created

FAISS Index Ready

LLM Loaded Successfully

Ask Question From PDF:
WHAT IS THE EDUCATION?

ANSWER:

Education Bachelor of Arts in Fashion Design,
University of Illinois, Chicago

Adobe Certified Expert, Adobe, Online

Associates Degree in Fashion Design,
University of Southern California, San Jacinto

Program Finished Successfully

Advantages

- Beginner-friendly AI project
- Practical implementation of RAG
- Fast document retrieval
- AI-powered question answering

Future Enhancements

- Streamlit Web App
- Multiple Question Support
- Advanced LLM Integration
- Cloud Deployment

Conclusion

This project demonstrates intelligent document understanding using NLP, vector databases, and Generative AI technologies.